Analytical Chemistry Cumulative Exam
February 8, 2008

Equations you might not know that would be helpful:

$$\alpha_{H2A} = \frac{[H^+]^2}{[H^+]^2 + [H^+]K_1 + K_1K_2}$$

$$\alpha_{HA-} = \frac{[H^+]K_1}{[H^+]^2 + [H^+]K_1 + K_1K_2}$$

$$\alpha_{A2-} = \frac{K_1K_2}{[H^+]^2 + [H^+]K_1 + K_1K_2}$$

Equations that should be part of your cumulative knowledge:
Ionic strength
Equilibrium Constant
Mass Balance
Charge Balance
"Themes" or similar equations to those listed above that have been provided to help jog your memory.

Instructions: **Please answer the following questions. Please make sure I can read your answers.**

1. (20 points) Explain why the pH of a buffer is nearly independent of acid/conjugate base or base/conjugate acid concentrations? In what cases does this independence of concentration diminish? Use acetic acid as an example, $pK_a = 4.75$.

2. (30 points) Citric acid $H_3(C_6H_5O_7)$ ["H_3(citrate)" or "H_3A"] is commonly used in biological buffer systems. For this question, you have made a sodium citrate buffer with a pH of 5.0 that is 0.1 F.

For citric acid: pK_{a1}: 3.12 pK_{a2}: 4.8 pK_{a3}: 6.4

a) What are the concentrations of the four different citrate species (forms) in this solution?

b) What is the ionic strength of this solution?

3. (20 points) Using mass and charge balance correctly, calculate the pH of a 1 μM HNO_3 solution. Assume nitric acid has been added to ultrapure water free of carbon dioxide.

4. (30 points) Detection of calcium, Ca^{2+}, is extremely important in biological research as it is a signaling agent. Of particular interest is detection of Ca^{2+} in cells. Fluorophores such as Fluo-AM with a K_d (dissociation constant) of 320 nM are frequently used in biology laboratories around the world to measure Ca^{2+}. Fluo-AM will penetrate the cell wall and bind to Ca^{2+} and this complex will fluoresce. Assume calcium concentrations inside cells never reach greater than 100 nM for this question. Calcium concentrations in the cell are determined using values of K_d, maximum fluorescence, minimum fluorescence, and measured fluorescence.

a) Based on the equilibrium constant value, how would you decide how much Fluo-AM to add to the cell system? Note that Fluo-AM also has some background fluorescence so you need to minimize this fluorescence and maximize the capability to detect calcium concentration fluctuations. Justify an answer for the concentration of Fluo-AM.

b) Maximum fluorescence can be obtained by adding an ionophore which forces large concentrations of calcium into the cell to a level not higher than the 100 nM described above. EDTA is commonly used to bind calcium to minimize the fluorescence. Assume for this question the 1:1 association constant between calcium and EDTA is $2 \times 10^8\ M^{-1}$. EDTA is the last thing to be added to the cell after the initial measurement is made (with the concentration of fluorophore you calculated above residing in the cell) and the maximum fluorescence is induced. How much EDTA should be added? Justify your answer.